Q1

December 2016: F70LA solutions

(a) $p_{78} = 1 - {}^{\cdot 028326}q_{78}$

$= \underline{0.971674}$

(b) ${}_4q_{63} = 1 - \frac{l_{67}}{l_{63}}$

$= 1 - \frac{9605.483}{9775.888}$

$= \underline{0.017431}$

(c) $\frac{l_{80} - l_{70}}{l_{70}} = \frac{7724.737 - 4118.693}{9392.621}$

$= \underline{0.383923}$

(d) $l_{79.7} = 0.7 l_{80} + 0.3 l_{79}$

$= 0.7 \times 7724.737 + 0.3 \times 7979.465$

$= 7801.155$

$2.7 p_{77} = \frac{l_{79.7}}{l_{77}} = \frac{7801.155}{8423.243} = \underline{0.926146}$

(e) $e^{-\mu} = p_{80}$

$\mu = -\ln(p_{80}) = -\ln(1 - 0.035882)$

$= \underline{0.036542}$

Q2

Males: q_x increases rapidly
then decreases
then increases again
so a "hump"

Females: similar pattern
but much smaller hump (almost none)
and lower overall levels

Explanations (?)

- illness increases over years?
- accidents more likely for males?
- poorer health for males?

[1 mark per sensible point]

Q3

$$P = \int_{t=0}^{4} (400{,}000 - 100{,}000t)\, v^t \, {}_tP_x \, \mu_{x+t} \, dt$$

where $v = 0.8$

$$\mu_{x+t} = B c^{x+t}$$

$${}_tP_x = g^{c^x (c^t - 1)}$$

$$g = e^{-B/\ln c} = 0.992637$$

t	S	v^t	${}_tP_x$	μ_{x+t}	Weight	Product
0	400K	1	1	0.01686	1	675
1	300K	0.8	0.9827	0.01804	4	17,022
2	200K	0.64	0.9645	0.01931	2	4,767
3	100K	0.512	0.9455	0.02066	4	4,000
4	0	0.4096	0.9255	0.02210	1	0
						32,534

$h = 2$

Total $\times h/6 = £10{,}845$

Q4

(a) Value at 60 of an annuity of £1 pa

$$= \bar{a}_{\overline{5}|} + \frac{D_{65}}{D_{60}} \bar{a}_{65} \qquad (1)$$

$$\bar{a}_{\overline{5}|} = a_{\overline{5}|} \times i/\delta = 4.4518 \times 1.019869$$

$$\bar{a}_{65} \approx \underset{12.276}{\bar{a}_{65}} - \tfrac{1}{2} = 11.776$$

$$\therefore (1) = 4.5403 + \frac{689.23}{882.85} \times 11.776 = 13.734$$

Value of pension at 40 (including 0.5%) $= 10{,}050 \times \underset{882.85}{D_{60}} / \underset{2052.54}{D_{[40]}} \times 13.734$

$$= £59{,}369$$

$$\bar{A}^{1}_{[40]:\overline{20}|} \approx 1.04^{\underset{0.4642}{\frac{1}{2}}} \left(A_{[40]:\overline{20}|} - \frac{D_{60}}{D_{[40]}} \right)$$

$$= 0.034779$$

$$P = 59369 + 10\%\,P + P\,\bar{A}^{1}_{[40]:\overline{20}|}$$

$$P = 59369 / (1 - 0.1 - 0.034779)$$

$$P = \underline{\underline{£68{,}617}}$$

Reserve at 1/1/2015

$$= 200 \times \left(10050 \underset{1366.51}{\frac{D_{60}^{882.85}}{D_{50}}} \times 13.734 + 68,617 \, \bar{A}'_{50:\overline{10}|} \right)$$

$$\bar{A}'_{50:\overline{10}|} \approx 1.04^{1/2} \left(\underset{.68024}{A^{1}_{50:\overline{10}|}} - \frac{D_{60}}{D_{50}} \right) = 0.034855$$

Reserve = £18,313,088

Amount held at 1/1/2016

$$= 18,313,088 \times 1.021 - 68617 \, (1.021)^{1.075}$$

$$= £18,627,968$$

Reserve at 1/1/2016

$$= 199 \times \left(10050 \underset{1310.75}{\frac{D_{60}^{882.85}}{D_{51}}} \times 13.734 + 68617 \, \bar{A}'_{51:\overline{9}|} \right)$$

$$\bar{A}_{51:\overline{9}|} \approx (1.04)^{1/2} \left(\underset{.070672}{A_{51:\overline{9}|}} - \frac{D_{60}}{D_{51}} \right) = 0.0338304$$

Reserve = £18,962,446

Loss = 18,962,446 − 18,627,968

= £334,479

Lower interest than expected (big loss);

deaths $\frac{1}{200} = 0.005$, $q_x = 0.0025$ (v. small profit, reserve > SA)

Q5

(a) Value of fixed part of death benefit $= 100,000 \left(A^{0.34204}_{32:28\overline{]}} - \frac{D_{60}}{D_{[32]}} \right) \times 1.04^{\frac{1}{2}}$

$= £3,017$

Value of increasing part of death benefit $= \left[5000 \left\{ (IA)^{7.14328}_{[32]} \right\} \right.$

$\left. - \frac{D^{882.85}_{60}}{D_{[32]}} \left(28 \times 5000 A^{0.45640}_{60} + 5000 (IA)^{8.36324}_{60} \right) \right.$

$\times 1.04^{\frac{1}{2}}$

$= £2740$

Value of maturity benefit $= 240,000 \times \frac{D_{60}}{D_{[32]}}$

$= £74,990$

Total benefits $= £80,747$

Value of net premiums $= 0.97 P \ddot{a}^{(4)}_{[32]:28\overline{]}} - 1000$

$\ddot{a}^{(4)}_{[32]:28\overline{]}} = \ddot{a}^{17.107}_{[32]:28\overline{]}} - \frac{3}{8}\left(1 - \frac{D_{60}}{D_{[37]}}\right)$

$= 16.849$

$\therefore \quad 0.97 P \times 16.849 - 1000 = 80747$

$P = £5001.76$

quarterly $P = 5001.76/4 = £1250.44$

Q5 (b) $A_{se} = 39$, + 5 year rating = 44

Value of benefits = $143\,500\ A^{4\%}_{44:\overline{21}|}$

$= 143\,500 \times 0.45253$

$= £64,945$

Value of premiums = $0.94 \times 5400\ \ddot{a}^{(4)}_{44:\overline{21}|}$

$\ddot{a}^{(4)}_{44:\overline{21}|} = \ddot{a}^{6\%}_{44:\overline{21}|} - \frac{3}{8}\left(1 - \frac{D_{65}}{D_{44}}\right)$ ($8821.262 / 9814.3354 \times \frac{1}{1.06}$)

$= 12.197 - \frac{3}{8}(1 - 0.89877)$

$= 11.921$

Value of premiums = £60,512

Reserve = £64,945 − £60512

= £4433

(c) 1) True: 4433 << 7 × 5400

2) Premium higher than calculated

3) Interest rate has increased leading to lower reserve

4) Future bonuses in reserving basis lower than in premium basis.

[accept other sensible suggestions]

Q6

$$\bar{A}'_x = \int_{t=0}^{\infty} v^t \, {}_tp'_x \, \mu'_{x+t} \, dt$$

Now ${}_tp'_x = \exp\left(-\int_0^t \mu'_{x+s} \, ds\right)$

$$= \exp\left(-\int_0^t \mu_{x+s} + \phi \, ds\right)$$

$$= \exp\left(-\int_0^t \mu_{x+s} \, ds \; \underbrace{-\int_0^t \phi \, ds}_{\phi t}\right)$$

$$= ({}_tp_x) \, e^{-\phi t}$$

$$\bar{A}'_x = \int_{t=0}^{\infty} e^{-(\delta+\phi)t} \, {}_tp_x \, (\mu_{x+t} + \phi) \, dt$$

$$= \int_{t=0}^{\infty} e^{-(\delta+\phi)t} \, {}_tp_x \, \mu_{x+t} \, dt + \phi \int_{t=0}^{\infty} e^{-(\delta+\phi)} \, {}_tp_x \, dt$$

$$\bar{A}'_x = \bar{A}_x + \phi \bar{a}_x$$

where both functions are calculated using a force of interest $= \delta + \phi$.